

Calculus of Variations

Euler's Equation, Standard Variational Problems, Geodesics,
Isoperimetric Problems, Higher-Order Functionals & the Rayleigh–Ritz Method

Digitized Lecture Notes – M.Tech (ASE), Advanced Engineering Mathematics

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1 Euler's Equation

1.1 Euler's Equation

A **necessary condition** for

$$I = \int_{x_1}^{x_2} f(x, y, y') dx$$

to be an **extremum** is that

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad (1)$$

This is called **Euler's equation**.

1.2 Other Forms of Euler's Equation

By considering f and $\frac{\partial f}{\partial y'}$ as a function of (x, y, y') , we get **two more relations** of Euler's equations. They are:

$$\frac{d}{dx} \left(f - y' \frac{\partial f}{\partial y'} \right) - \frac{\partial f}{\partial x} = 0 \quad (2)$$

and

$$\frac{\partial f}{\partial y} - \frac{\partial^2 f}{\partial x \partial y'} - y' \frac{\partial^2 f}{\partial y \partial y'} - y'' \frac{\partial^2 f}{\partial y'^2} = 0 \quad (3)$$

2 Solution of Euler's Equation

2.1 Solution of Euler's Equation — Case (i)

Case (i): When f is independent of x

$$\Rightarrow \text{we get } \frac{\partial f}{\partial x} = 0$$

$\therefore$ By Euler's eqn. (2), we have

$$\frac{d}{dx} \left(f - y' \frac{\partial f}{\partial y'} \right) = 0$$

$$\Rightarrow f - y' \frac{\partial f}{\partial y'} = \text{constant}$$

$$\Rightarrow f = y' \frac{\partial f}{\partial y'} \quad \Rightarrow \quad \frac{\partial f}{f} = \frac{\partial y'}{y'}$$

Solving the above eqn., we get the solution for f .

2.2 Solution of Euler's Equation — Case (ii)

Case (ii): Suppose f is independent of y

$$\Rightarrow \frac{\partial f}{\partial y} = 0$$

∴ By Euler's eqn. (1), we get

$$-\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0 \quad \text{i.e.} \quad \boxed{\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0}$$

$$\Rightarrow \frac{\partial f}{\partial y'} = \text{constant}$$

Integrating, we get its solution for f .

2.3 Solution of Euler's Equation — Case (iii)

Case (iii): Suppose f is independent of y'

$$\Rightarrow \frac{\partial f}{\partial y'} = 0$$

∴ By Euler's eqn. (1),

$$\frac{\partial f}{\partial y} = 0$$

$$\int \Rightarrow \text{we get its solution for } f.$$

2.4 Solution of Euler's Equation — Case (iv)

Case (iv): Suppose f is independent of both x and y

$$\begin{aligned} \Rightarrow \frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0, \quad \frac{\partial^2 f}{\partial y' \partial x} = 0 \quad \text{and} \quad \frac{\partial^2 f}{\partial y' \partial y} = 0 \\ \Rightarrow \frac{\partial^2 f}{\partial x \partial y'} = 0 \quad \text{and} \quad \frac{\partial^2 f}{\partial y \partial y'} = 0 \end{aligned}$$

∴ By Euler's eqn. (3), we get

$$y'' \frac{\partial^2 f}{\partial y'^2} = 0$$

If $\frac{\partial^2 f}{\partial y'^2} \neq 0$, then we get

$$y'' = 0 \quad \Rightarrow \quad \frac{d^2 y}{dx^2} = 0$$

Integrating,

$$\int \Rightarrow \frac{dy}{dx} = a, \text{ constant}$$

Integrating again,

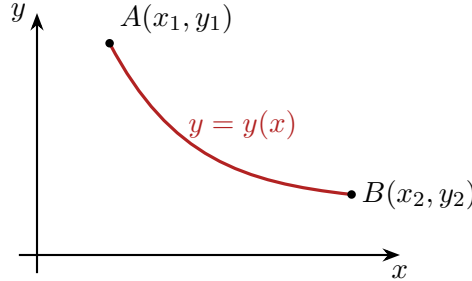
$$y = ax + b, \quad a, b \text{ constants (a straight line).}$$

3 Standard Variational Problem 1: Shortest Plane Curve

3.1 Standard Variational Problems — Problem 1

Problem 1. Find the **shortest smooth plane curve** joining two distinct points in the plane.

Solution:



Let $y = y(x)$ be the shortest smooth plane curve joining the two distinct points $A(x_1, y_1)$ and $B(x_2, y_2)$ of the xy plane.

3.2 Problem 1 — Setting up the Functional

Since $y(x)$ is the shortest smooth plane curve, then the *arc length joining its points A and B is smallest (minimum)*.

i.e.

$$I = \int_{x_1}^{x_2} \sqrt{1 + y'^2} \, dx \text{ is minimum} \quad (\because \text{by arc length formula})$$

which is of the form

$$\int_{x_1}^{x_2} f(x, y, y') \, dx$$

$$\Rightarrow f(x, y, y') = \sqrt{1 + y'^2}$$

Now, by Euler's eqn. for extremum of I ,

3.3 Problem 1 — Applying Euler's Equation

We have

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad (1)$$

But $f(x, y, y') = \sqrt{1 + y'^2}$

$$\Rightarrow \frac{\partial f}{\partial y} = 0$$

and

$$\begin{aligned} \frac{\partial f}{\partial y'} &= \frac{\partial}{\partial y'} \left(\sqrt{1 + y'^2} \right) = \frac{\partial}{\partial y'} (1 + y'^2)^{1/2} \\ &= \frac{1}{2} (1 + y'^2)^{\frac{1}{2}-1} (2y') = (1 + y'^2)^{-1/2} y' \end{aligned}$$

3.4 Problem 1 — Solving

$$\Rightarrow \frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{1+y'^2}}$$

Substituting these in (1):

$$0 - \frac{d}{dx} \left(\frac{y'}{\sqrt{1+y'^2}} \right) = 0 \quad \Rightarrow \quad \frac{d}{dx} \left(\frac{y'}{\sqrt{1+y'^2}} \right) = 0$$

Integrating,

$$\int \Rightarrow \frac{y'}{\sqrt{1+y'^2}} = \text{constant} = k \text{ (say)}$$

$$\Rightarrow y' = k\sqrt{1+y'^2} \quad \Rightarrow \quad y'^2 = k^2(1+y'^2) = k^2 + k^2 y'^2$$

3.5 Problem 1 — Conclusion: Straight Line

$$y'^2(1-k^2) = k^2 \Rightarrow y'^2 = \frac{k^2}{1-k^2} \Rightarrow y' = \frac{k}{\sqrt{1-k^2}} \Rightarrow \frac{dy}{dx} = \frac{k}{\sqrt{1-k^2}}$$

Integrating,

$$\int \Rightarrow \int dy = \int \frac{k}{\sqrt{1-k^2}} dx \Rightarrow y = \frac{k}{\sqrt{1-k^2}} x + c$$

which is of the form $y = mx + c$, where $m = \frac{k}{\sqrt{1-k^2}}$ — it is the eqn. of a **straight line**.

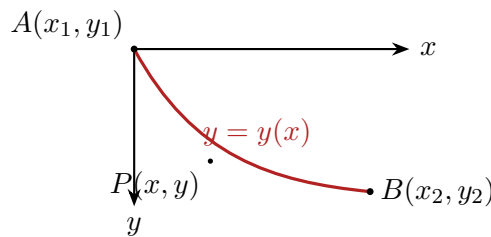
$\therefore$ Straight line is the only possible smooth curve joining the two points A and B in the shortest distance.

4 Brachistochrone (Shortest Time) Problem

4.1 Problem 2: Brachistochrone (Shortest Time) Problem

Problem. Determine the plane curve down which a particle will slide **without friction** from the point $A(x_1, y_1)$ to $B(x_2, y_2)$ in the **shortest time**.

Solution: Let us assume that the positive direction of the y axis is vertically downward ($\because$ particle falls down). We have the fig.:



4.2 Brachistochrone — Velocity of the Particle

Here the particle slides down from the point $A(x_1, y_1)$ to $B(x_2, y_2)$ along the curve $y = y(x)$. Let $P(x, y)$ be the position of the particle on the curve $y = y(x)$ at any time t .

Hence the speed/velocity (v) of the particle sliding along the curve is given by

$$v = \sqrt{2g(y - y^*)} \quad (1)$$

where

$$y^* = y_1 - \frac{v_1^2}{2g} \quad (\text{motion of the particle}) \quad (2)$$

where g – acceleration due to gravity, v_1 – initial velocity (speed).

4.3 Brachistochrone — Simplification

Choose the point A as origin, then $(x_1, y_1) = (0, 0)$

$$\Rightarrow x_1 = 0 ; \boxed{y_1 = 0}$$

Moreover the initial velocity $\boxed{v_1 = 0}$ (at rest).

$$\therefore (1) \ \& \ (2) \Rightarrow v = \sqrt{2gy} \quad (\because \text{sub. } v_1 = 0, y_1 = 0 \text{ in } (2) \text{ and } (1))$$

$$\Rightarrow \frac{ds}{dt} = \sqrt{2gy} \quad (\because v \text{ is velocity})$$

$$\Rightarrow \frac{dt}{ds} = \frac{1}{\sqrt{2gy}} \quad \Rightarrow dt = \frac{1}{\sqrt{2gy}} ds$$

4.4 Brachistochrone — Time Functional

Integrating,

$$\begin{aligned} \int \Rightarrow t &= \int \frac{1}{\sqrt{2gy}} ds = \frac{1}{\sqrt{2g}} \int \frac{1}{\sqrt{y}} ds \\ &= \frac{1}{\sqrt{2g}} \int_{x_1}^{x_2} \frac{1}{\sqrt{y}} \sqrt{1 + y'^2} dx \quad (\because ds \text{ in terms of } dx) \\ t &= \frac{1}{\sqrt{2g}} \int_0^{x_2} \frac{\sqrt{1 + y'^2}}{\sqrt{y}} dx \quad (\because A \text{ is origin}) \\ t &= \int f(x, y, y') dx \text{ with minimum time,} \end{aligned}$$

$$\text{where } f(x, y, y') = \frac{\sqrt{1 + y'^2}}{\sqrt{y}} \quad (3)$$

4.5 Brachistochrone — Applying Euler's Equation

i.e. by solving the above integral, we have to find the curve $y = y(x)$ with the restriction of shortest time. $\therefore$ We can use Euler's eqn.

In $f(x, y, y') = \frac{\sqrt{1 + y'^2}}{\sqrt{y}}$, f seems **independent of x** .

$\therefore$ By the case of Euler's eqn. for independent of x , we get

$$f - y' \frac{\partial f}{\partial y'} = \text{constant} \quad \text{i.e.} \quad f - y' \frac{\partial f}{\partial y'} = k \quad (4)$$

Now from (3),

$$\begin{aligned}\frac{\partial f}{\partial y'} &= \frac{1}{\sqrt{y}} \frac{\partial}{\partial y'} \left(\sqrt{1+y'^2} \right) = \frac{1}{\sqrt{y}} \cdot \frac{1}{2} (1+y'^2)^{\frac{1}{2}-1} (2y') \\ &= \frac{y'}{\sqrt{y} \sqrt{1+y'^2}}, \quad \text{sub. this in (4)}\end{aligned}$$

4.6 Brachistochrone — Reducing the Equation

$$\begin{aligned}(4) \Rightarrow \frac{\sqrt{1+y'^2}}{\sqrt{y}} - y' \left(\frac{y'}{\sqrt{y} \sqrt{1+y'^2}} \right) &= k \\ \Rightarrow \frac{(1+y'^2) - y'^2}{\sqrt{y} \sqrt{1+y'^2}} = k &\Rightarrow \frac{1}{\sqrt{y} \sqrt{1+y'^2}} = k \\ \Rightarrow \sqrt{y} \sqrt{1+y'^2} &= \frac{1}{k}\end{aligned}$$

Squaring,

$$\Rightarrow y (1+y'^2) = \frac{1}{k^2} = c, \text{ a constant}$$

$$\text{i.e. } y (1+y'^2) = c \quad (5)$$

4.7 Brachistochrone — Parametrisation $y' = \cot \theta$

Let $y' = \cot \theta$. Then (5) $\Rightarrow$

$$\begin{aligned}y &= \frac{c}{1+y'^2} = \frac{c}{1+\cot^2 \theta} = \frac{c}{1+\frac{\cos^2 \theta}{\sin^2 \theta}} = \frac{c \sin^2 \theta}{\sin^2 \theta + \cos^2 \theta} \\ &= c \sin^2 \theta = c \left(\frac{1 - \cos 2\theta}{2} \right)\end{aligned}$$

$$\Rightarrow y = \frac{c}{2} (1 - \cos 2\theta) \quad (6)$$

4.8 Brachistochrone — To Get x in Terms of θ

Already we have $y' = \cot \theta$

$$\Rightarrow \frac{dy}{dx} = \cot \theta \Rightarrow \frac{dx}{dy} = \tan \theta \Rightarrow dx = \tan \theta dy$$

$$\int \Rightarrow x = \int \tan \theta dy \quad (7)$$

Already $y = \frac{c}{2} (1 - \cos 2\theta)$

$$\Rightarrow dy = \frac{c}{2} (0 - (-\sin 2\theta)(2)) d\theta \Rightarrow \boxed{dy = c \sin 2\theta d\theta}$$

4.9 Brachistochrone — Integrating for x

$$\begin{aligned} (7) \Rightarrow x &= \int \tan \theta \, dy = \int \tan \theta \, c \sin 2\theta \, d\theta = c \int \frac{\sin \theta}{\cos \theta} 2 \sin \theta \cos \theta \, d\theta \\ &= 2c \int \sin^2 \theta \, d\theta = 2c \int \left(\frac{1 - \cos 2\theta}{2} \right) d\theta = c \left[\theta - \frac{\sin 2\theta}{2} \right] \end{aligned}$$

$$\Rightarrow x = \frac{c}{2} [2\theta - \sin 2\theta] \quad (8)$$

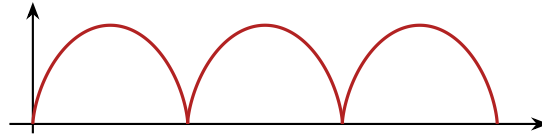
$\therefore$ By (6) & (8), we have the parametric form of the curve.

4.10 Brachistochrone — The Cycloid

$\therefore$ The required curve $y = y(x)$ is

$$x = \frac{c}{2} (2\theta - \sin 2\theta) \quad \text{and} \quad y = \frac{c}{2} (1 - \cos 2\theta)$$

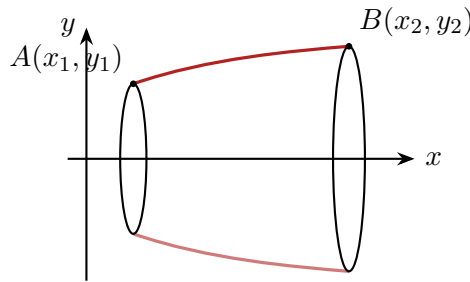
and it is the eqn. of the **cycloid curve**:



5 Minimal Surface Area

5.1 Problem 3: Minimal Surface Area

Problem. Find the curve $C : y = f(x)$ passing through two points $A(x_1, y_1)$ and $B(x_2, y_2)$ such that the rotation of the curve C about the x -axis generates a surface of revolution having **minimum surface area**.



5.2 Minimal Surface Area — The Functional

Solution: The surface area generated by revolution (rotation of the curve) of C is defined by

$$S = \int_{AB} 2\pi y \, ds = \int_A^B 2\pi y \sqrt{1 + y'^2} \, dx$$

$$S = 2\pi \int_A^B y \sqrt{1 + y'^2} \, dx \quad (1)$$

which is of the form $S = 2\pi \int_A^B f(x, y, y') dx$.

$\therefore$ To get the curve $y = f(x)$ by minimizing the above surface area, we have to extremize the above functional and get its curve $y = f(x)$. $\therefore$ We can use Euler's eqn. to extremize it.

5.3 Minimal Surface Area — Applying Euler's Equation

Here $f(x, y, y') = y\sqrt{1+y'^2}$ and is independent of x .

$\therefore$ By Euler's eqn. for independent of x condn., we have

$$f - y' \frac{\partial f}{\partial y'} = k, \text{ a constant} \quad (2)$$

Now $f = y\sqrt{1+y'^2} = y(1+y'^2)^{1/2}$

$$\Rightarrow \frac{\partial f}{\partial y'} = y \cdot \frac{1}{2} (1+y'^2)^{\frac{1}{2}-1} (2y') = y y' (1+y'^2)^{-1/2}$$

$$\frac{\partial f}{\partial y'} = \frac{y y'}{\sqrt{1+y'^2}}, \quad \text{sub. in (2)}$$

5.4 Minimal Surface Area — Reduction

$$\begin{aligned} (2) \Rightarrow y\sqrt{1+y'^2} - y' \left(\frac{y y'}{\sqrt{1+y'^2}} \right) &= k \\ \Rightarrow \frac{y(1+y'^2) - y y'^2}{\sqrt{1+y'^2}} &= k \quad \Rightarrow \quad y + y y'^2 - y y'^2 = k\sqrt{1+y'^2} \\ \Rightarrow y &= k\sqrt{1+y'^2} \end{aligned}$$

Put $y' = \sinh t$:

$$\Rightarrow y = k\sqrt{1+\sinh^2 t} = k\sqrt{\cosh^2 t} \quad (\because \cosh^2 t - \sinh^2 t = 1)$$

$$y = k \cosh t$$

5.5 Minimal Surface Area — To Get $x = x(t)$

Already $y' = \sinh t$

$$\Rightarrow \frac{dy}{dx} = \sinh t \Rightarrow \frac{dx}{dy} = \frac{1}{\sinh t} \Rightarrow dx = \frac{dy}{\sinh t}$$

Already $y = k \cosh t \Rightarrow dy = k \sinh t dt$. Integrating,

$$\int \Rightarrow x = \int \frac{dy}{\sinh t} = \int \frac{k \sinh t dt}{\sinh t} = k \int dt$$

$$x = kt + a, \quad \text{where } a \text{ is constant}$$

5.6 Minimal Surface Area — The Catenary

i.e. the eqn. of the curve in terms of parametric form is

$$x = kt + a, \quad y = k \cosh t$$

Sub $t = \frac{x-a}{k}$ in y , then

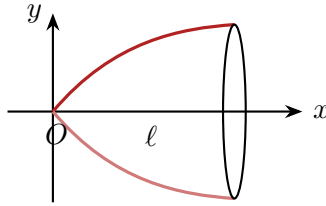
$$y = k \cosh\left(\frac{x-a}{k}\right)$$

is the required curve eqn. (**Catenary**).

6 Solid of Revolution with Least Resistance

6.1 Problem 4: Solid of Revolution with Least Resistance

Problem. Determine the shape of a solid of revolution moving in a flow of gas with **least resistance**.



Solution: In a flow of a gas, the total resistance experienced by the body is given by

$$I(y(x)) = 4\pi\rho v^2 \int_0^\ell y y'^3 dx \quad (1)$$

6.2 Least Resistance — Setting Up

with boundary condns.

$$y(0) = 0 \quad \text{and} \quad y(\ell) = k,$$

where ρ – density of the gas, v – velocity of the gas relative to the solid.

Here we have to identify the shape of the solid by minimising the total resistance given by the eqn. (1).

In (1), $f(x, y, y') = y y'^3$ and is **independent of x** .

$\therefore$ By Euler's eqn., for independent of x condn., we have

$$f - y' \frac{\partial f}{\partial y'} = c, \text{ a constant} \quad (2)$$

6.3 Least Resistance — Reduction

Now $f(x, y, y') = y y'^3$

$$\frac{\partial f}{\partial y'} = y \cdot 3y'^2 = 3y y'^2$$

$$\therefore (2) \Rightarrow y y'^3 - y' (3y y'^2) = c \Rightarrow y y'^3 - 3y y'^3 = c \Rightarrow -2y y'^3 = c$$

$$\Rightarrow y y'^3 = c_1, \quad \text{where } c_1 = \frac{-c}{2}$$

$$\Rightarrow y'^3 = \frac{c_1}{y} \Rightarrow y' = \left(\frac{c_1}{y} \right)^{1/3} \Rightarrow \frac{dy}{dx} = \frac{c_1^{1/3}}{y^{1/3}}$$

6.4 Least Resistance — Integration

$$\begin{aligned} \Rightarrow y^{1/3} dy &= c_1^{1/3} dx \quad \Rightarrow \quad y^{1/3} dy = k_1 dx \quad \text{where } k_1 = c_1^{1/3} \\ \int \Rightarrow \int y^{1/3} dy &= \int k_1 dx \quad \Rightarrow \quad \frac{y^{\frac{1}{3}+1}}{\frac{1}{3}+1} = k_1 x + k_2 \quad \Rightarrow \quad \frac{y^{4/3}}{4/3} = k_1 x + k_2 \\ \Rightarrow y^{4/3} &= \frac{4}{3} k_1 x + \frac{4}{3} k_2 \end{aligned}$$

$$y^{4/3} = m_1 x + m_2 \quad (3)$$

where $m_1 = \frac{4}{3} k_1$, $m_2 = \frac{4}{3} k_2$.

6.5 Least Resistance — Applying Boundary Conditions

Now apply initial and boundary condns. $y(0) = 0$ and $y(\ell) = k$ in (3):

When $y(0) = 0$ in (3): $0 = 0 + m_2 \Rightarrow \boxed{m_2 = 0}$

$$\therefore \text{ soln (3) } \Rightarrow y^{4/3} = m_1 x \quad (4)$$

When $y(\ell) = k$, (4) $\Rightarrow k^{4/3} = m_1 \ell \Rightarrow \boxed{m_1 = \frac{k^{4/3}}{\ell}}$

$$\therefore (4) \Rightarrow y^{4/3} = \frac{k^{4/3}}{\ell} x \quad \Rightarrow \quad y = \frac{k}{\ell^{3/4}} x^{3/4}$$

$$y = k \left(\frac{x}{\ell} \right)^{3/4}$$

7 Extremals of Functionals

7.1 Problem Involving All x, y, y'

Problem. Find the extremal (or) complete solution of the functional

$$\int_{x_1}^{x_2} (y^2 - y'^2 - 2y \cosh x) dx$$

Solution: Given the functional

$$I = \int_{x_1}^{x_2} (y^2 - y'^2 - 2y \cosh x) dx$$

and $f(x, y, y') = y^2 - y'^2 - 2y \cosh x$.

To extremize the above functional, consider Euler's eqn.

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad (1)$$

7.2 Deriving the ODE

Now $f = y^2 - y'^2 - 2y \cosh x$

$$\frac{\partial f}{\partial y} = 2y - 2 \cosh x \quad \text{and} \quad \frac{\partial f}{\partial y'} = -2y', \quad \text{sub. in (1)}$$

$$\begin{aligned} (1) &\Rightarrow (2y - 2 \cosh x) - \frac{d}{dx} (-2y') = 0 \\ &\Rightarrow 2y - 2 \cosh x + 2y'' = 0 \quad \Rightarrow \quad 2y'' + 2y = 2 \cosh x \end{aligned}$$

$$\Rightarrow y'' + y = \cosh x \quad (2)$$

i.e. $\frac{d^2 y}{dx^2} + y = \cosh x$, i.e. $(D^2 + 1)y = \cosh x$ —(3), where $D = \frac{d}{dx}$.

7.3 Solving: Complementary Function

Soln of (3) is $y = \text{C.F} + \text{P.I.}$

To find C.F: A.E of (3) is

$$\begin{aligned} m^2 + 1 &= 0 \quad (\because D \text{ by } m) \\ m^2 &= -1 \quad \Rightarrow \quad m = \pm i = \alpha \pm i\beta \\ \therefore \text{C.F} &= e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x) = e^{0x} (c_1 \cos x + c_2 \sin x) \end{aligned}$$

$$\text{C.F} = c_1 \cos x + c_2 \sin x$$

7.4 Solving: Particular Integral

To find P.I:

$$\begin{aligned} \text{P.I} &= \frac{1}{D^2 + 1} \cosh x = \frac{1}{D^2 + 1} \left(\frac{e^x + e^{-x}}{2} \right) \\ &= \frac{1}{2} \left\{ \frac{1}{D^2 + 1} e^x + \frac{1}{D^2 + 1} e^{-x} \right\} \quad (\text{Replace } D \text{ by } 1, D \text{ by } -1) \\ &= \frac{1}{2} \left\{ \frac{1}{2} e^x + \frac{1}{2} e^{-x} \right\} = \frac{1}{2} \left\{ \frac{e^x + e^{-x}}{2} \right\} \end{aligned}$$

$$\text{P.I} = \frac{1}{2} \cosh x$$

7.5 Complete Solution

∴ Soln of the given eqn. is

$$y = \text{C.F} + \text{P.I}$$

$$y = c_1 \cos x + c_2 \sin x + \frac{1}{2} \cosh x$$

8 Test for Extremum

8.1 Problem: Test for Extremum of the Functional

Problem. Test for extremum of the functional

$$I = \int_0^{\pi/2} (y'^2 - y^2) dx, \quad \text{given that } \boxed{y(0) = 0} \quad \text{and} \quad \boxed{y(\pi/2) = 1}$$

Solution: Given

$$I = \int_0^{\pi/2} (y'^2 - y^2) dx \quad \Rightarrow \quad f(x, y, y') = y'^2 - y^2$$

To extremize the gn. functional, apply Euler's eqn.

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad (1)$$

8.2 Test for Extremum — The ODE

Now $f = y'^2 - y^2$

$$\Rightarrow \frac{\partial f}{\partial y} = -2y \quad \left| \quad \frac{\partial f}{\partial y'} = 2y' \right.$$

$$\therefore (1) \Rightarrow -2y - \frac{d}{dx} (2y') = 0 \quad \Rightarrow \quad -2y - 2y'' = 0$$

$$\Rightarrow y'' + y = 0 \quad (2)$$

which is a 2nd order ODE.

To get soln (2): (2) is $(D^2 + 1)y = 0$, where $D = \frac{d}{dx}$. A.E of the above eqn. is

$$m^2 + 1 = 0 \Rightarrow m^2 = -1 \Rightarrow m = \pm i \Rightarrow \text{C.F} = c_1 \cos x + c_2 \sin x$$

8.3 Test for Extremum — Boundary Conditions

Since RHS = 0 $\Rightarrow$ C.F is the general soln.

$$\Rightarrow \text{Soln of (2) is } y = c_1 \cos x + c_2 \sin x \quad (3)$$

Now apply the boundary condn. $y(0) = 0$ in (3):

$$\Rightarrow 0 = c_1(1) + c_2(0) \Rightarrow \boxed{c_1 = 0} \quad \therefore \text{soln (3)} \Rightarrow y = c_2 \sin x \quad (4)$$

Now apply another boundary condn. $y(\pi/2) = 1$ in (4):

$$\text{then (4)} \Rightarrow 1 = c_2 \sin \frac{\pi}{2} \Rightarrow \boxed{c_2 = 1}$$

$\therefore$ soln (4) $\Rightarrow y = \sin x$ is the soln of the gn. functional.

9 Invalid Variational Problem

9.1 Invalid Variational Problem

Problem. Test for an extremum of the functional

$$I = \int_0^1 (xy^2 + y^2 - 2y^2y') dx, \text{ given that } \boxed{y(0) = 1 \text{ and } y(1) = 2}$$

Solution: Given $f = xy^2 + y^2 - 2y^2y'$. Then

$$\begin{aligned} \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) &= 0 \Rightarrow (2xy + 2y - 4yy') - \frac{d}{dx} (-2y^2) = 0 \\ &\Rightarrow 2xy + 2y - 4yy' + 2(2yy') = 0 \end{aligned}$$

$$\text{i.e. } 2xy + 2y - 4yy' + 4yy' = 0 \quad \Rightarrow \quad xy = -y \Rightarrow 1 = -x$$

$$y \neq 0 \Rightarrow x = -1$$

9.2 Invalid Variational Problem — Conclusion

i.e. the obtained curve $x = -1$ **does not satisfy** the gn. boundary condns. $y(0) = 1$ and $y(1) = 2$.

$\therefore$ The gn. problem is an **invalid variational problem**, i.e. we cannot extremize the gn. functional based on the gn. condns.

10 Exercises

10.1 Exercise: Find the Extremals

Ex. Find the extremals of the following functionals:

1. $\int_0^1 (y'^2 + 12xy) dx$, gn. $y(0) = 0$ and $y(1) = 1$.
2. $\int_{x_1}^{x_2} (y^2 + y'^2 - 2y \sin x) dx$.
3. $\int_{x_1}^{x_2} (y^2 + 2xy y') dx$, gn. $y(x_1) = y_1$ and $y(x_2) = y_2$.

11 Geodesics

11.1 Geodesics

A variational problem involving **conditional extremum** is known as finding **geodesics** on a surface.

A **geodesic** on a surface is a curve along which the *distance between any two points of the surface is minimum*.

11.2 Geodesics — Problem 1

Problem 1. Show that geodesics on a plane are straight lines.

Solution: Let $y = y(x)$ be the geodesics on the xy plane and let $A(x_1, y_1)$ and $B(x_2, y_2)$ be two points on the plane.

$\Rightarrow$ Geodesics $y(x)$ connecting the pts A and B with minimum distance.

W.k.t, in a curve $y = y(x)$, the distance b/w any two pts is measured by its arc length:

$$s = \int_{x_1}^{x_2} \sqrt{1 + y'^2} dx$$

11.3 Geodesics on a Plane are Straight Lines

$\therefore$ Geodesics $y = y(x)$ on the xy plane is the curve with minimum arc length s . So we have to minimize s and get the curve $y = y(x)$.

To do this, here $f(x, y, y') = \sqrt{1 + y'^2}$, and by Euler's eqn. for independent of x and y , we get

$$y'' = 0 \quad \text{i.e.} \quad \frac{d^2 y}{dx^2} = 0 \quad \int \Rightarrow \frac{dy}{dx} = a$$

Integrating again,

$$y = ax + b$$

it's the eqn. of a straight line. $\Rightarrow$ Geodesics on a plane is a straight line.

12 Geodesics on a Right Circular Cylinder

12.1 Geodesics on a Right Circular Cylinder

Problem

Find the geodesics on a right circular cylinder of radius a .

Solution. Consider the **cylindrical coordinates**: for a circular cylinder,

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

Now the arc length of the two elements of a right circular cylinder of radius r can be given by

$$ds^2 = (dr)^2 + (r d\theta)^2 + dz^2.$$

Since $r = a$ (constant) $\Rightarrow dr = 0$.

$$\begin{aligned} \Rightarrow ds^2 &= 0 + a^2 (d\theta)^2 + (dz)^2 \\ &= a^2 d\theta^2 + (dz)^2 \\ &= (d\theta)^2 \left(a^2 + \left(\frac{dz}{d\theta} \right)^2 \right) \\ (ds)^2 &= (d\theta)^2 (a^2 + z'^2), \quad \text{where } z' = \frac{dz}{d\theta}. \end{aligned}$$

$$\Rightarrow ds = \sqrt{a^2 + z'^2} d\theta \quad \Rightarrow \quad s = \int_{\theta_1}^{\theta_2} \sqrt{a^2 + z'^2} d\theta \quad (1)$$

Key idea

The geodesic on the cylinder is nothing but the curve with **minimum** s . Hence we can use **Euler's equation** on $f(\theta, z, z') = \sqrt{a^2 + z'^2}$ of (1).

Here $f = \sqrt{a^2 + z'^2}$ is **independent of z and θ** .

$\therefore$ By Euler's equation, for f independent of z and θ , we get

$$z'' = 0 \quad \implies \quad \frac{d^2 z}{d\theta^2} = 0.$$

Integrating: $\frac{dz}{d\theta} = a$ (a constant).

Integrating again:

$$z = a\theta + b$$

Conclusion

$z = a\theta + b$ is a **circular helix** — the geodesic on a right circular cylinder.

13 Isoperimetric Problems

13.1 Isoperimetric Problems

Find $y = y(x)$ by extremising the functional

$$I = \int_{x_1}^{x_2} f(x, y, y') dx$$

with respect to the condition

$$J = \int_{x_1}^{x_2} g(x, y, y') dx \text{ (a given constant).}$$

Lagrangian multiplier method

Consider

$$H(x, y, y') = f(x, y, y') + \lambda g(x, y, y')$$

where λ is the **Lagrangian multiplier**. Then Euler's equation for $H(x, y, y')$ is

$$\frac{\partial H}{\partial y} - \frac{d}{dx} \left(\frac{\partial H}{\partial y'} \right) = 0$$

13.2 Isoperimetric Problems — Parametric Form

Suppose both f and g are given in terms of the parametric equations, i.e. $x = x(t)$ and $y = y(t)$. Then

- $f(x, y, y')$ can be written in the form $f(x, y, \dot{x}, \dot{y})$, where $\dot{x} = \frac{dx}{dt}$ and $\dot{y} = \frac{dy}{dt}$;
- similarly $g(x, y, y')$ can be written in the form $g(x, y, \dot{x}, \dot{y})$;
- and $H = f + \lambda g$, i.e. $H = H(x, y, \dot{x}, \dot{y})$.

Hence the respective Euler's equations are

$$\frac{\partial H}{\partial x} - \frac{d}{dt} \left(\frac{\partial H}{\partial \dot{x}} \right) = 0 \quad \text{and} \quad \frac{\partial H}{\partial y} - \frac{d}{dt} \left(\frac{\partial H}{\partial \dot{y}} \right) = 0$$

(the parametric analogue of $\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$ with $x = x(t)$, $y = y(t)$).

13.3 Problem 1: A Constrained Extremal

Problem

Find the extremal of the functional $I = \int_0^\pi (y'^2 - y^2) dx$ with boundary conditions $y(0) = 0$ and $y(\pi) = 1$, subject to the constraint $\int_0^\pi y dx = 1$.

Solution. Here we have to extremise $\int_0^\pi (y'^2 - y^2) dx$. Hence

$$f(x, y, y') = y'^2 - y^2$$

$$\text{and the condition is } \int_0^\pi y dx = 1 \Rightarrow g(x, y, y') = y$$

∴ By Lagrangian multiplier, we consider

$$H = f + \lambda g = (y'^2 - y^2) + \lambda y.$$

$$\Rightarrow \frac{\partial H}{\partial y} = -2y + \lambda \quad \text{and} \quad \frac{\partial H}{\partial y'} = 2y'.$$

∴ By Euler's equation for H :

$$\frac{\partial H}{\partial y} - \frac{d}{dx} \left(\frac{\partial H}{\partial y'} \right) = 0$$

$$\begin{aligned} \Rightarrow -2y + \lambda - \frac{d}{dx} (2y') &= 0 \\ \Rightarrow -2y + \lambda - 2y'' &= 0 \\ \Rightarrow 2y'' + 2y &= \lambda \\ \Rightarrow y'' + y &= \frac{\lambda}{2} \end{aligned}$$

$$\text{i.e. } y'' + y = k \quad \text{where } k = \frac{\lambda}{2}; \text{ (1) is a 2}^{\text{nd}} \text{ order ODE.} \quad (1)$$

Now the solution of (1) is given by $y = \text{C.F.} + \text{P.I.}$

C.F.: Auxiliary equation of (1) is

$$m^2 + 1 = 0 \Rightarrow m^2 = -1 \Rightarrow m = \pm i = \alpha \pm i\beta \Rightarrow \alpha = 0, \beta = 1$$

$$\Rightarrow \text{C.F.} = c_1 \cos x + c_2 \sin x$$

P.I.:

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + 1} k \quad (\because \text{by (1)}) \\ &= k \frac{1}{D^2 + 1} e^{0x} \quad \text{Replace } D \text{ by } 0 \\ &= k e^{0x} \Rightarrow \text{P.I.} = k \end{aligned}$$

$$\therefore \text{Solution is } y = \text{C.F.} + \text{P.I.} \Rightarrow y = c_1 \cos x + c_2 \sin x + k \quad (2)$$

Now apply the boundary condition $y(0) = 0$ in the solution (2):

$$\begin{aligned} (2) \Rightarrow 0 &= c_1 \cos(0) + c_2 \sin(0) + k \\ 0 &= c_1 + 0 + k \\ \Rightarrow c_1 + k &= 0 \quad \Rightarrow c_1 = -k \end{aligned} \quad (3)$$

Now apply $y(\pi) = 1$ in (2):

$$\begin{aligned} (2) \Rightarrow 1 &= c_1 \cos \pi + c_2 \sin \pi + k \\ 1 &= c_1(-1) + 0 + k \\ \Rightarrow -c_1 + k &= 1 \end{aligned} \quad (4)$$

$$(3) + (4) \Rightarrow k = \frac{1}{2} \quad \therefore c_1 = -\frac{1}{2}$$

$\therefore$ Solution (2) is

$$y = c_1 \cos x + c_2 \sin x + k \Rightarrow y = -\frac{1}{2} \cos x + c_2 \sin x + \frac{1}{2} \quad (5)$$

By the given constraint, we have $\int_0^\pi y dx = 1$:

$$\begin{aligned} \Rightarrow \int_0^\pi \left(-\frac{1}{2} \cos x + c_2 \sin x + \frac{1}{2} \right) dx &= 1 \\ \Rightarrow -\frac{1}{2} \left(\sin x \right)_0^\pi + c_2 \left(-\cos x \right)_0^\pi + \frac{1}{2} \left(x \right)_0^\pi &= 1 \\ \Rightarrow -\frac{1}{2}(0 - 0) - c_2((-1) - 1) + \frac{1}{2}(\pi - 0) &= 1 \\ \Rightarrow 2c_2 = 1 - \frac{\pi}{2} = \frac{2 - \pi}{2} \quad \Rightarrow c_2 = \frac{2 - \pi}{4} \end{aligned}$$

13.4 Problem 1: Final Solution

$\therefore$ Solution (5) becomes

Required solution

$$y = -\frac{1}{2} \cos x + \left(\frac{2 - \pi}{4} \right) \sin x + \frac{1}{2}$$

13.5 Problem 2: Constraint $\int y^2 dx = 1$

Problem

Find the extremal of $I = \int_0^\pi y'^2 dx$ subject to the condition $\int_0^\pi y^2 dx = 1$, satisfying the boundary conditions $y(0) = 0$ and $y(\pi) = 1$.

Solution. Here $f = y'^2$, $g = y^2$, so $H = f + \lambda g = y'^2 + \lambda y^2$. Euler's equation for H gives

$$\begin{aligned} 2(y'' - \lambda y) &= 0 \\ y'' - \lambda y &= 0 \\ \text{i.e. } (D^2 - \lambda)y &= 0 \\ (D^2 - k^2)y &= 0, \quad \text{where } \lambda = k^2. \end{aligned} \quad (1)$$

Since R.H.S. = 0, the C.F. is itself the solution.

C.F. of (1): Auxiliary equation of (1) is

$$m^2 - k^2 = 0 \Rightarrow m^2 = k^2 \Rightarrow m = \pm k$$

$$\Rightarrow \text{C.F.} = c_1 e^{kx} + c_2 e^{-kx} \quad \therefore \text{Solution is } y = c_1 e^{kx} + c_2 e^{-kx} \quad (2)$$

Now apply the boundary conditions ($c_1 = ?$, $c_2 = ?$).

Apply $y(0) = 0$ in (2):

$$0 = c_1 + c_2 \quad \Rightarrow \quad c_1 = -c_2 \quad (\text{or}) \quad c_2 = -c_1$$

Apply $y(\pi) = 1$ in (2):

$$1 = c_1 e^{k\pi} + c_2 e^{-k\pi}$$

$$1 = c_1 e^{k\pi} - c_1 e^{-k\pi} \quad (\because c_2 = -c_1)$$

$$\Rightarrow 1 = c_1 (e^{k\pi} - e^{-k\pi})$$

$$\Rightarrow 1 = c_1 2 \sinh k\pi \quad \left(\because \sinh k\pi = \frac{e^{k\pi} - e^{-k\pi}}{2} \right)$$

Result

$$c_1 = \frac{1}{2 \sinh k\pi}, \quad c_2 = -c_1 = -\frac{1}{2 \sinh k\pi}$$

$$\Rightarrow y = \frac{e^{kx} - e^{-kx}}{2 \sinh k\pi} = \frac{\sinh kx}{\sinh k\pi}.$$

13.6 Problem (Parametric Form): Closed Curve of Maximum Area

Problem

Determine a closed curve C of given length which encloses the **maximum area**.

Solution. We know that the area of a closed curve C can be given by

$$\text{Area} = \frac{1}{2} \oint_C (x dy - y dx).$$

The above integral can also be written as

$$\text{Area} = \frac{1}{2} \oint_C (x\dot{y} - y\dot{x}) dt \quad (1)$$

if $x = x(t)$ and $y = y(t)$, where $\dot{x} = \frac{dx}{dt}$ and $\dot{y} = \frac{dy}{dt}$.

Further, the length of the closed curve C connecting two points t_1 and t_2 is given by

$$\text{Length} = \int_{t_1}^{t_2} \sqrt{\dot{x}^2 + \dot{y}^2} dt \quad (2)$$

The problem is to extremise the area (1) with respect to the length (2). Here we have

$$f = \frac{1}{2} (x\dot{y} - y\dot{x}) \quad \text{and} \quad g = \sqrt{\dot{x}^2 + \dot{y}^2}.$$

∴ We have $H = f + \lambda g$:

$$H = \frac{1}{2}(x\dot{y} - y\dot{x}) + \lambda\sqrt{\dot{x}^2 + \dot{y}^2} \quad (3)$$

Hence the Euler's equations under parametric form are

$$\frac{\partial H}{\partial x} - \frac{d}{dt}\left(\frac{\partial H}{\partial \dot{x}}\right) = 0 \quad \text{and} \quad \frac{\partial H}{\partial y} - \frac{d}{dt}\left(\frac{\partial H}{\partial \dot{y}}\right) = 0.$$

Now from (3):

$$\frac{\partial H}{\partial x} = \frac{\dot{y}}{2} \quad \text{and} \quad \frac{\partial H}{\partial y} = -\frac{\dot{x}}{2}.$$

Also

$$\begin{aligned} \frac{\partial H}{\partial \dot{x}} &= -\frac{y}{2} + \lambda \frac{1}{2} (\dot{x}^2 + \dot{y}^2)^{-1/2} (2\dot{x}) &\Rightarrow \quad \frac{\partial H}{\partial \dot{x}} &= -\frac{y}{2} + \frac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \\ \frac{\partial H}{\partial \dot{y}} &= \frac{x}{2} + \lambda \frac{1}{2} (\dot{x}^2 + \dot{y}^2)^{-1/2} (2\dot{y}) &\Rightarrow \quad \frac{\partial H}{\partial \dot{y}} &= \frac{x}{2} + \frac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \end{aligned}$$

∴ Euler's equation $\frac{\partial H}{\partial x} - \frac{d}{dt} \frac{\partial H}{\partial \dot{x}} = 0$ gives

$$\begin{aligned} \Rightarrow \quad \frac{\dot{y}}{2} - \frac{d}{dt} \left(-\frac{y}{2} + \frac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right) &= 0 \\ \Rightarrow \quad \frac{d}{dt} \left(-\frac{y}{2} + \frac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right) &= \frac{1}{2} \frac{dy}{dt} \end{aligned}$$

Integrating w.r.t. t on both sides:

$$\begin{aligned} -\frac{y}{2} + \frac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} &= \frac{y}{2} + c_1 \\ \text{i.e.} \quad y &= \frac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} + c_1 \end{aligned} \quad (4)$$

Similarly, the other Euler equation $\frac{\partial H}{\partial y} - \frac{d}{dt} \left(\frac{\partial H}{\partial \dot{y}} \right) = 0$ gives

$$\begin{aligned} \Rightarrow \quad -\frac{\dot{x}}{2} - \frac{d}{dt} \left(\frac{x}{2} + \frac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right) &= 0 \\ \Rightarrow \quad -\frac{1}{2} \frac{dx}{dt} = \frac{d}{dt} \left(\frac{x}{2} + \frac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right) \end{aligned}$$

Integrating w.r.t. t on both sides, we get

$$\begin{aligned} -\frac{x}{2} &= \frac{x}{2} + \frac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} + c_2 \\ \Rightarrow \quad x &= -\frac{\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} + c_2 \end{aligned} \quad (5)$$

From (4): $y - c_1 = \frac{\lambda \dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}}$ and from (5): $x - c_2 = \frac{-\lambda \dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}}.$

Squaring and adding, we then have

$$(x - c_2)^2 + (y - c_1)^2 = \frac{\lambda^2 \dot{y}^2}{\dot{x}^2 + \dot{y}^2} + \frac{\lambda^2 \dot{x}^2}{\dot{x}^2 + \dot{y}^2} = \frac{\lambda^2 (\dot{x}^2 + \dot{y}^2)}{\dot{x}^2 + \dot{y}^2}$$

13.7 Maximum Area: Conclusion

Result

$$(x - c_2)^2 + (y - c_1)^2 = \lambda^2$$

is a **circle** with centre (c_2, c_1) and radius λ . $\Rightarrow$ The **circle** is the required closed curve having **maximum area** with respect to the given length.

13.8 Exercise (Isoperimetric)

Exercise

Show that the extremal of the isoperimetric problem

$$I_1 = \int_{x_1}^{x_2} y'^2 dx \quad \text{subject to the condition}$$

$$I_2 = \int_{x_1}^{x_2} y dx = k \quad (\text{a constant})$$

is a **parabola**. Also determine the equation of the parabola when it passes through the points $(1, 3)$ and $(4, 24)$ and also $k = 36$.

Hint (from notes). $f = y'^2$ and $g = y \Rightarrow H = f + \lambda g$.

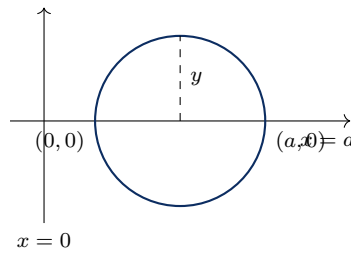
13.9 Problem: The Sphere as a Solid of Revolution

Problem

Prove that the sphere is the solid of figure of revolution which, for a given surface area, has minimum volume.

Solution. For a given surface, its surface area is given by

$$I = \int_{x=0}^a 2\pi y ds \quad \Rightarrow \quad I = \int_0^a 2\pi y \sqrt{1 + y'^2} dx \quad (1)$$



Hence the volume of the respective surface is given by

$$V = \int_0^a \pi y^2 dx \quad (2)$$

Here we have to minimize V subject to the surface area (1):

$$\therefore f = \pi y^2 \quad \text{and} \quad g = 2\pi y \sqrt{1 + y'^2}.$$

$$\therefore H = f + \lambda g \quad \Rightarrow \quad H = \pi y^2 + \lambda 2\pi y \sqrt{1 + y'^2} \quad (3)$$

Then by Euler's equation, for H independent of x , we have (Beltrami identity)

$$H - y' \frac{\partial H}{\partial y'} = k, \text{ a constant} \quad \left(\text{cf. } f - y' \frac{\partial f}{\partial y'} = \text{constant} \right)$$

$$\begin{aligned} & \left(\pi y^2 + 2\pi\lambda y \sqrt{1 + y'^2} \right) - y' \left(0 + 2\pi\lambda y \cdot \frac{1}{2} (1 + y'^2)^{-1/2} (2y') \right) = k \\ \Rightarrow & \pi y^2 + 2\pi\lambda y \sqrt{1 + y'^2} - \frac{2\pi\lambda y y'^2}{\sqrt{1 + y'^2}} = k \\ \Rightarrow & \pi y^2 + \frac{2\pi\lambda y (1 + y'^2) - 2\pi\lambda y y'^2}{\sqrt{1 + y'^2}} = k \\ \Rightarrow & \pi y^2 + \frac{2\pi\lambda y}{\sqrt{1 + y'^2}} = k \end{aligned} \tag{4}$$

Since the curve is passing through both $(0,0)$ and $(a,0)$, we have $y = 0$ at these two points. Substituting $y = 0$ in (4):

$$k = 0$$

$$\begin{aligned} \therefore (4) \Rightarrow \quad \pi y^2 &= -\frac{2\pi\lambda y}{\sqrt{1 + y'^2}} \quad \Rightarrow \quad y = \frac{-2\lambda}{\sqrt{1 + y'^2}} \\ &\Rightarrow \quad y \sqrt{1 + y'^2} = -2\lambda. \end{aligned}$$

Squaring:

$$\begin{aligned} y^2 (1 + y'^2) &= 4\lambda^2 \quad \Rightarrow \quad y^2 + y^2 y'^2 = 4\lambda^2 \\ \Rightarrow \quad y^2 y'^2 &= 4\lambda^2 - y^2 \quad \Rightarrow \quad y'^2 = \frac{4\lambda^2 - y^2}{y^2} \\ \Rightarrow \quad y' &= \frac{\sqrt{4\lambda^2 - y^2}}{y} \quad \Rightarrow \quad \frac{dy}{dx} = \frac{\sqrt{4\lambda^2 - y^2}}{y} \\ \Rightarrow \quad \frac{y dy}{\sqrt{4\lambda^2 - y^2}} &= dx \quad \Rightarrow \quad \int dx = \int \frac{y dy}{\sqrt{4\lambda^2 - y^2}} \end{aligned}$$

On the R.H.S. let $t = 4\lambda^2 - y^2 \Rightarrow dt = -2y dy$:

$$\begin{aligned} \Rightarrow \quad x &= \int \frac{-dt/2}{\sqrt{t}} \\ \Rightarrow \quad x &= -\frac{1}{2} \int t^{-1/2} dt = -\frac{1}{2} \left(\frac{t^{-1/2+1}}{-\frac{1}{2}+1} \right) + c = -\frac{1}{2} \cdot \frac{t^{1/2}}{\frac{1}{2}} + c \\ &= -\sqrt{t} + c \\ x &= c - \sqrt{4\lambda^2 - y^2} \quad \Rightarrow \quad (x - c) = -\sqrt{4\lambda^2 - y^2} \end{aligned}$$

Squaring:

$$(x - c)^2 = 4\lambda^2 - y^2 \quad \Rightarrow \quad (x - c)^2 + y^2 = 4\lambda^2$$

Result

$$(x - c)^2 + y^2 = (2\lambda)^2$$

is the equation of the **circle**. Hence the **sphere** is the solid of figure of revolution of the above circle. ■

14 Functionals Involving Several Dependent Variables

14.1 Functionals Involving Several Dependent Variables

So far we have extremized the functionals such as $f(x, y, y')$ using Euler's equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0.$$

Now we are going to extremize the functionals that involve several variables such as

$$f(x, y, y', y''), \quad f(x, y, y', y'', y'''), \quad \text{etc.}$$

Euler's equation for $f(x, y, y', y'')$

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial f}{\partial y''} \right) = 0 \quad (1)$$

14.2 Ostrogradsky Equation

Similarly, for the extremization of the functional $f(x, y, y', y'', y''')$, the Euler's equation is

Euler's equation for $f(x, y, y', y'', y''')$

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial f}{\partial y''} \right) - \frac{d^3}{dx^3} \left(\frac{\partial f}{\partial y'''} \right) = 0 \quad (2)$$

... and so on for other functionals — the **Ostrogradsky equation**.

14.3 Problem: Show that $y = \sin x$ Extremizes the Functional

Problem

Show that the curve which extremizes the functional $I = \int_0^{\pi/4} (y''^2 - y^2 + x^2) dx$ under the conditions $y(0) = 0$, $y'(0) = 1$, $y(\pi/4) = y'(\pi/4) = \frac{1}{\sqrt{2}}$ is $y = \sin x$.

Solution. Given $I = \int_0^{\pi/4} (y''^2 - y^2 + x^2) dx$. To extremize the above functional, we have

$$\begin{aligned} f(x, y, y', y'') &= y''^2 - y^2 + x^2 \\ \Rightarrow \quad \frac{\partial f}{\partial y} &= -2y, \quad \frac{\partial f}{\partial y'} = 0, \quad \frac{\partial f}{\partial y''} = 2y''. \end{aligned}$$

The corresponding equation is

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial f}{\partial y''} \right) = 0$$

$$\begin{aligned}
\Rightarrow & -2y - 0 + \frac{d^2}{dx^2} (2y'') = 0 \\
\Rightarrow & -2y + 2 \frac{d}{dx} \left(\frac{d}{dx} (y'') \right) = 0 \\
\Rightarrow & -2y + 2 \frac{d}{dx} (y''') = 0 \\
\Rightarrow & -2y + 2y^{(iv)} = 0 \quad \Rightarrow \quad 2y^{(iv)} = 2y \\
\Rightarrow & \boxed{y^{(iv)} - y = 0} \tag{1}
\end{aligned}$$

i.e. $(D^4 - 1)y = 0$. $\therefore$ The auxiliary equation is

$$\begin{aligned}
m^4 - 1 = 0 & \Rightarrow (m^2)^2 - (1)^2 = 0 \\
& \Rightarrow (m^2 + 1)(m^2 - 1) = 0 \\
& \Rightarrow m^2 = -1 \text{ (or) } m^2 = 1 \\
& \Rightarrow m = \pm i \text{ (or) } m = \pm 1
\end{aligned}$$

$$\Rightarrow \text{Solution is } \boxed{y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x} \tag{2}$$

Apply $y(0) = 0$ in (2):

$$\Rightarrow 0 = c_1 + c_2 + c_3 \tag{3}$$

Apply $y(\pi/4) = \frac{1}{\sqrt{2}}$ in (2):

$$\Rightarrow \frac{1}{\sqrt{2}} = c_1 e^{\pi/4} + c_2 e^{-\pi/4} + c_3 \frac{1}{\sqrt{2}} + c_4 \frac{1}{\sqrt{2}} \tag{4}$$

From (2):

$$y' = c_1 e^x - c_2 e^{-x} - c_3 \sin x + c_4 \cos x \tag{5}$$

Apply $y'(0) = 1$ in (5):

$$\Rightarrow 1 = c_1 - c_2 + c_4 \tag{6}$$

Apply $y'(\pi/4) = \frac{1}{\sqrt{2}}$ in (5):

$$\Rightarrow \frac{1}{\sqrt{2}} = c_1 e^{\pi/4} - c_2 e^{-\pi/4} - \frac{c_3}{\sqrt{2}} + \frac{c_4}{\sqrt{2}} \tag{7}$$

$$(3) + (6) \Rightarrow 1 = 2c_1 + c_3 + c_4 \tag{8}$$

Now (4) + (7):

$$\begin{aligned}
\Rightarrow \frac{2}{\sqrt{2}} &= 2c_1 e^{\pi/4} + \frac{2c_4}{\sqrt{2}} \quad \Rightarrow \quad c_1 e^{\pi/4} + \frac{c_4}{\sqrt{2}} = \frac{1}{\sqrt{2}} \\
\Rightarrow c_1 e^{\pi/4} &= \frac{1}{\sqrt{2}} (1 - c_4) \quad \Rightarrow \quad \boxed{c_1 = \frac{e^{-\pi/4}}{\sqrt{2}} (1 - c_4)} \tag{9}
\end{aligned}$$

Substitute (9) in (8): $2c_1 + c_3 + c_4 = 1$

$$\begin{aligned}
\Rightarrow c_3 &= 1 - c_4 - 2c_1 = 1 - c_4 - 2 \frac{e^{-\pi/4}}{\sqrt{2}} (1 - c_4) \\
\Rightarrow \boxed{c_3} &= (1 - c_4) \left(1 - e^{-\pi/4} \sqrt{2} \right) \tag{10}
\end{aligned}$$

Now substitute (9) & (10) in (3): $c_1 + c_2 + c_3 = 0$

$$\begin{aligned}\Rightarrow c_2 + \frac{e^{-\pi/4}}{\sqrt{2}} c_4 (2-1) &= \frac{e^{-\pi/4}}{\sqrt{2}} (2-1) - 1 \\ \Rightarrow c_2 + \frac{e^{-\pi/4}}{\sqrt{2}} c_4 &= \frac{e^{-\pi/4}}{\sqrt{2}} - 1\end{aligned}\quad (11)$$

Substitute (9) & (10) in (6): $c_1 - c_2 + c_4 = 1$

$$\begin{aligned}\Rightarrow \frac{e^{-\pi/4}}{\sqrt{2}} (1 - c_4) - c_2 + c_4 &= 1 \\ \Rightarrow c_4 \left(1 - \frac{e^{-\pi/4}}{\sqrt{2}}\right) - c_2 &= 1 - \frac{e^{-\pi/4}}{\sqrt{2}}\end{aligned}\quad (12)$$

Adding (11) + (12):

$$\Rightarrow c_4 \left(\frac{e^{-\pi/4}}{\sqrt{2}} + 1 - \frac{e^{-\pi/4}}{\sqrt{2}} \right) = \frac{e^{-\pi/4}}{\sqrt{2}} - 1 + 1 - \frac{e^{-\pi/4}}{\sqrt{2}} + 1$$

We get

$$c_4 = 1 \quad \therefore \quad c_1 = 0, \quad c_3 = 0, \quad c_2 = 0$$

14.4 $y = \sin x$: Final Solution & Exercise

Required solution (from (2))

$$y = \sin x$$

Exercise

Find the extremal of the functional $\int_0^{\pi/2} (y''^2 - y^2 + x^2) dx$ satisfying the conditions

$$y(0) = 1; \quad y'(0) = 0; \quad y(\pi/2) = 0; \quad y'(\pi/2) = -1.$$

15 Unit II: Direct Methods for Solving Boundary Value Problems

15.1 Unit II: Direct Methods for Solving Boundary Value Problems

In this section, we will see the **boundary value problems** on the Euler's differential equations of some variational problems given by the functional

$$I = \int_{x_1}^{x_2} f(x, y, y') dx. \quad (1)$$

15.2 Unit II: The Rayleigh–Ritz Method

Rayleigh–Ritz method

In this method, we will construct a **trial solution** $\bar{y}(x)$ for the given boundary value problem using the **Lagrangian interpolation method**. Substitute this $\bar{y}(x)$ in (1) (i.e. in the

functional), i.e.

$$I = \int_{x_1}^{x_2} f(x, \bar{y}, \bar{y}') dx,$$

Determining the constants

Then find out the required arbitrary constants $c_1, c_2, \dots$ with the help of the equations

$$\frac{\partial I}{\partial c_1} = 0, \quad \frac{\partial I}{\partial c_2} = 0, \dots$$

and hence we get the solution of the problem.

15.3 Rayleigh–Ritz Problem 1

Problem 1

Solve the boundary value problem $y'' - y + x = 0$ with $y(0) = 0$ and $y(1) = 0$, by extremizing the functional $I = \int_0^1 (2xy - y^2 - y'^2) dx$ by the Rayleigh–Ritz method.

Solution. Given the functional $I = \int_0^1 (2xy - y^2 - y'^2) dx$

$$\Rightarrow f = 2xy - y^2 - y'^2 \quad \Rightarrow \quad \frac{\partial f}{\partial y} = 2x - 2y, \quad \frac{\partial f}{\partial y'} = -2y'.$$

$\therefore$ By Euler's equation, we have

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

$$\Rightarrow 2x - 2y - \frac{d}{dx} (-2y') = 0$$

$$\Rightarrow 2x - 2y + 2y'' = 0$$

$$\Rightarrow y'' + x - y = 0 \quad (2)$$

which is the required boundary value problem, with $y(0) = 0$ and $y(1) = 0$.

Let us construct a trial solution for the problem (2) through the Lagrangian interpolation method. For

x	0 (x_0)	$\frac{1}{2}$ (x_1)	1 (x_2)
y	0 (y_0)	c (y_1)	0 (y_2)

$\therefore$ By Lagrange's interpolation formula,

$$\bar{y}(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

$$\Rightarrow \bar{y}(x) = \frac{(x - \frac{1}{2})(x - 1)}{(0 - \frac{1}{2})(0 - 1)} (0) + \frac{(x - 0)(x - 1)}{(\frac{1}{2} - 0)(\frac{1}{2} - 1)} (c) + \frac{(x - 0)(x - \frac{1}{2})}{(1 - 0)(1 - \frac{1}{2})} (0)$$

$$\begin{aligned}
\Rightarrow \bar{y}(x) &= 0 + \frac{x(x-1)}{\frac{1}{2}(-\frac{1}{2})}(c) + 0 \\
&= -4cx(x-1) \\
&= 4cx(1-x)
\end{aligned}$$

$$\bar{y}(x) = c_1 x(1-x) = c_1 (x - x^2)$$

$$\text{where } c_1 = 4c$$

Further $\bar{y}' = c_1(1-2x)$.

Now substitute $\bar{y}$ and $\bar{y}'$ in the given functional:

$$\begin{aligned}
I &= \int_0^1 (2x\bar{y} - \bar{y}^2 - \bar{y}'^2) dx \\
&= \int_0^1 \left[2x c_1 (x - x^2) - c_1^2 (x - x^2)^2 - c_1^2 (1 - 2x)^2 \right] dx \\
&= \int_0^1 \left[2c_1 (x^2 - x^3) - c_1^2 (x^2 + x^4 - 2x^3) - c_1^2 (1 + 4x^2 - 4x) \right] dx \\
&= 2c_1 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 - c_1^2 \left[\frac{x^3}{3} + \frac{x^5}{5} - \frac{2x^4}{4} \right]_0^1 - c_1^2 \left[x + \frac{4x^3}{3} - \frac{4x^2}{2} \right]_0^1 \\
&= 2c_1 \left(\left(\frac{1}{3} - \frac{1}{4} \right) - 0 \right) - c_1^2 \left(\left(\frac{1}{3} + \frac{1}{5} - \frac{1}{2} \right) - 0 \right) - c_1^2 \left(\left(1 + \frac{4}{3} - 2 \right) - 0 \right) \\
&= \frac{c_1}{6} - c_1^2 \left(\frac{2}{3} + \frac{1}{5} - \frac{3}{2} \right) = \frac{c_1}{6} - c_1^2 \left(\frac{20 + 6 - 45}{30} \right) = \frac{c_1}{6} - c_1^2 \left(\frac{-19}{30} \right) \\
\Rightarrow I &= \frac{c_1}{6} + \frac{19c_1^2}{30}
\end{aligned}$$

To get c_1 , consider $\frac{\partial I}{\partial c_1} = 0$:

$$\frac{1}{6} + \frac{38c_1}{30} = 0 \quad \Rightarrow \quad \frac{38c_1}{30} = -\frac{1}{6} \quad \Rightarrow \quad c_1 = -\frac{5}{38}$$

15.4 Rayleigh–Ritz Problem 1: Final Solution

$\therefore$ The solution is given by (the trial solution)

$$y(x) = c_1 x(1-x) = -\frac{5}{38} x(1-x)$$

Required solution of the given boundary value problem

$$y(x) = \frac{5}{38} x(x-1)$$

15.5 Rayleigh–Ritz Problem 2

Problem 2

Solve the boundary value problem $x^2 y'^3 - x^3 y'^2 y'' = 0$ by the Rayleigh–Ritz method, extremizing the functional $\int_1^2 \frac{x^3}{y'^2} dx$ with $y(1) = 0$ and $y(2) = 3$.

Solution. Given the functional $I = \int_1^2 \frac{x^3}{y'^2} dx$

$$\Rightarrow f = \frac{x^3}{y'^2} \quad \text{and } f \text{ is independent of } y.$$

$\therefore$ The respective Euler's equation is

$$\frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \quad \Rightarrow \quad \frac{d}{dx} \left(\frac{y'^2 \cdot 0 - x^3 (2y')}{y'^4} \right) = 0$$

$$\Rightarrow \frac{d}{dx} \left(-\frac{2x^3 y'}{y'^4} \right) = 0 \quad \Rightarrow \quad \frac{d}{dx} \left(-\frac{2x^3}{y'^3} \right) = 0$$

$$\Rightarrow \frac{(y'^3)(-6x^2) - (-2x^3)(3y'^2)y''}{(y'^3)^2} = 0$$

$$\Rightarrow -6x^2 y'^3 + 6x^3 y'^2 y'' = 0$$

$$\Rightarrow -x^2 y'^3 + x^3 y'^2 y'' = 0 \quad (2)$$

which is the given boundary value problem.

Aside: $x^3 y'^2 y'' = x^2 y'^3 \Rightarrow xy'' = y' \Rightarrow x \frac{d^2 y}{dx^2} - \frac{dy}{dx} = 0 \Rightarrow (x^2 D^2 - xD)y = 0$.

To get the trial solution $\bar{y}(x)$ for the above equation (2), we have

x	1 (x_0)	1.5 (x_1)	2 (x_2)
y	0 (y_0)	c (y_1)	3 (y_2)

$\therefore$ By Lagrange's interpolation formula,

$$\begin{aligned} \bar{y}(x) &= \frac{(x-1.5)(x-2)}{(1-1.5)(1-2)}(0) + \frac{(x-1)(x-2)}{(1.5-1)(1.5-2)}(c) \\ &\quad + \frac{(x-1)(x-1.5)}{(2-1)(2-1.5)}(3) \end{aligned}$$

$$= 0 + \frac{x^2 - 3x + 2}{(0.5)(-0.5)}(c) + \frac{x^2 - 2.5x + 1.5}{1(0.5)}(3)$$

$$= (x^2 - 3x + 2) \left(\frac{c}{-0.25} \right) + (x^2 - 2.5x + 1.5) \left(\frac{3}{0.5} \right)$$

$$= (x^2 - 3x + 2)(-4c) + (x^2 - 2.5x + 1.5)(6)$$

$$\bar{y}(x) = x^2(-4c + 6) + x(12c - 15) + (-8c + 9)$$

$$\Rightarrow \bar{y}'(x) = 2x(-4c + 6) + (12c - 15) = x(12 - 8c) + (12c - 15)$$

$$\Rightarrow \quad \bar{y}'(x) = c_1 x + c_2 \quad \text{where } c_1 = 12 - 8c \text{ and } c_2 = 12c - 15.$$

Substituting in I :

$$I = \int_1^2 \frac{x^3}{\bar{y}'^2} dx = \int_1^2 \frac{x^3}{(c_1 x + c_2)^2} dx = \int_1^2 \frac{x^3}{c_1^2 x^2 + 2c_1 c_2 x + c_2^2} dx$$

$$\text{Let } t = c_1 x + c_2 \Rightarrow x = \frac{t - c_2}{c_1}, \quad dx = \frac{dt}{c_1}; \quad \text{when } x = 1, t = c_1 + c_2; \text{ when } x = 2, t = 2c_1 + c_2.$$

$$\begin{aligned} I &= \int_{c_1+c_2}^{2c_1+c_2} \frac{\left(\frac{t-c_2}{c_1}\right)^3}{t^2} \frac{dt}{c_1} = \int_{c_1+c_2}^{2c_1+c_2} \frac{1}{c_1^4} \frac{(t-c_2)^3}{t^2} dt \\ &= \frac{1}{c_1^4} \int_{c_1+c_2}^{2c_1+c_2} \left(\frac{t^3 - c_2^3 + 3tc_2^2 - 3t^2 c_2}{t^2} \right) dt \\ &= \frac{1}{c_1^4} \int_{c_1+c_2}^{2c_1+c_2} \left(t - c_2^3 t^{-2} + \frac{3c_2^2}{t} - 3c_2 \right) dt \\ &= \frac{1}{c_1^4} \left[\left(\frac{t^2}{2} \right)_{c_1+c_2}^{2c_1+c_2} - c_2^3 \left(\frac{t^{-1}}{-1} \right)_{c_1+c_2}^{2c_1+c_2} + 3c_2^2 \left(\log t \right)_{c_1+c_2}^{2c_1+c_2} - 3c_2 \left(t \right)_{c_1+c_2}^{2c_1+c_2} \right] \end{aligned}$$

= **lengthy calculations** (as noted in the class notes – the evaluation of $\partial I/\partial c_1 = 0$, $\partial I/\partial c_2 = 0$ becomes tedious).

Therefore (continuation of Problem 2)

Instead, the obtained differential equation (2) is solved *exactly* on the next slides.

15.6 Rayleigh–Ritz Problem 2 — Continuation: Exact Solution

Obtained differential equation is

$$-x^2 y'^3 + x^3 y'^2 y'' = 0 \quad (2)$$

$$\begin{aligned} \text{i.e. } x^3 y'^2 y'' &= x^2 y'^3 & \Rightarrow & \quad x y'' = y' \\ \text{i.e. } x y'' - y' &= 0 & \xrightarrow{\times \text{ by } x} & \quad x^2 y'' - x y' = 0 \end{aligned}$$

$$\text{i.e. } (x^2 D^2 - x D) y = 0 \quad (3)$$

To convert (3) into a differential equation with **constant coefficients**, put

$$\begin{aligned} x &= e^z \text{ (or) } z = \log x; & \text{then} \\ x D &= \theta, & x^2 D^2 = \theta(\theta - 1), & \text{where } \theta = \frac{d}{dz}. \end{aligned}$$

$$\begin{aligned} \therefore (3) \Rightarrow (\theta(\theta - 1) - \theta) y &= 0 \Rightarrow (\theta^2 - \theta - \theta) y = 0 \\ &\Rightarrow (\theta^2 - 2\theta) y = 0 \end{aligned} \quad (4)$$

which is a differential equation with constant coefficients.

Solution of (4): Auxiliary equation of (4) is

$$m^2 - 2m = 0 \Rightarrow m(m - 2) = 0 \Rightarrow m = 0, 2.$$

$\therefore$ Solution of (4) is

$$y = c_1 e^{0z} + c_2 e^{2z} \Rightarrow y = c_1 + c_2 e^{2z}.$$

$\therefore$ Solution of (3) is obtained by putting $x = e^z$ (or) $z = \log x$:

$$y = c_1 + c_2 x^2$$

i.e. $y(x) = c_1 + c_2 x^2$ is the required solution of the given boundary value problem. (5)

Now apply $y(1) = 0$ in (5):

$$(5) \Rightarrow c_1 + c_2 = 0 \quad (6)$$

Now apply $y(2) = 3$ in (5):

$$(5) \Rightarrow 3 = c_1 + 4c_2 \quad (7)$$

Solving (6) & (7): $(7) - (6) \Rightarrow 3c_2 = 3$

$$c_2 = 1$$

$$c_1 = -1$$

Required solution

$$\therefore (5) \Rightarrow y(x) = -1 + x^2 \implies y(x) = x^2 - 1$$

15.7 Rayleigh–Ritz Problem 3

Problem 3

Solve the boundary value problem $y'' - 6x = 0$, by extremizing the functional $\int_0^1 (y'^2 + 12xy) dx$, given $y(0) = 0$ and $y(1) = 1$.

Solution. Given the functional

$$I = \int_0^1 (y'^2 + 12xy) dx \quad (1)$$

$$\Rightarrow f = y'^2 + 12xy \Rightarrow \frac{\partial f}{\partial y} = 12x, \quad \frac{\partial f}{\partial y'} = 2y'.$$

Then by Euler's equation, we have

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

$$\Rightarrow 12x - \frac{d}{dx} (2y') = 0$$

$$\Rightarrow 12x - 2y'' = 0$$

$$\Rightarrow 2y'' = 12x$$

$$\Rightarrow y'' - 6x = 0 \quad (2)$$

which is the required boundary value problem, with $y(0) = 0$ and $y(1) = 1$.

Equation (2) is a simple 2nd order ODE:

$$y'' = 6x$$

Integrating:

$$y' = \frac{6x^2}{2} + c_1 \quad \Rightarrow \quad y' = 3x^2 + c_1$$

Integrating again:

$$y = x^3 + c_1x + c_2 \quad \Rightarrow \quad y(x) = x^3 + c_1x + c_2 \quad (3)$$

which is its solution.

Now apply $y(0) = 0$ in (3) $\Rightarrow$ $0 = c_2$

15.8 Rayleigh–Ritz Problem 3: Final Solution

$\therefore$ Solution (3) $\Rightarrow$

$$y(x) = x^3 + c_1x \quad (4)$$

is the updated solution.

Now apply $y(1) = 1$:

$$(4) \Rightarrow 1 = 1 + c_1 \quad \Rightarrow \quad c_1 = 0$$

Required solution

$$\therefore (4) \Rightarrow y(x) = x^3$$

Next topic: Fourier Transform (continued in the next set of notes).